

Two-Regime Liquidity Recovery After a Perpetual Futures Liquidation Cascade:

Evidence from Hyperliquid and the October 10, 2025 Event

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Working draft — 2026-04-23

Abstract

We document the inside-market response of Hyperliquid — the largest decentralized perpetual futures venue — to the October 10, 2025 system-wide liquidation cascade. Using the venue's public L2 order-book archive, we construct a minute-level panel spanning BTC, ETH, SOL, XRP, and DOGE perpetual contracts across a 92-day window centered on the event. Three findings emerge. First, the cascade was a tightly compressed intraday event: median quoted spread pooled across assets reached 9.4 basis points at hour 21 UTC and the 95th percentile reached 87.5 basis points, with intraday recovery well under way by end of day. Second, across the 23-day post-event window prior to a separate market-wide selloff beginning November 3, 2025, quoted spreads on all five contracts returned to within 1.2 times pre-event levels and three of five contracts returned to within 1.1 times. Third, inside-quote depth contracted persistently over the same window, with four of five contracts showing further depth contraction from days 1-7 to days 17-23 post-event, ending between 66 percent and 79 percent of pre-event levels. We interpret these results as evidence that market-maker response to a perpetual futures tail event is two-regime: pricing recovers within weeks while inventory commitment persistently contracts. The pattern has implications for how execution quality and venue resilience should be measured: spread-only metrics systematically understate the post-event impairment of inside-market liquidity.

Keywords: decentralized exchange, market microstructure, perpetual futures, liquidation cascades, market resilience, order-book depth.

JEL classification: G14, G15, G23.

1. Introduction

Perpetual futures are the dominant derivative class in cryptocurrency markets, with global notional volumes that have exceeded sixty trillion US dollars annually. Until recently, all meaningful liquidity in this segment resided on centralized exchanges. Hyperliquid, launched in 2023 and scaled materially through 2024 and 2025, is the first decentralized venue to attain comparable order-book liquidity. Its market share of decentralized perpetual volume reached approximately

60-65 percent by mid-2025, and its single-venue daily volume regularly exceeds five billion US dollars. The protocol's distinguishing feature is a consensus-layer ordering rule: within each block, the validator software is required to process cancel orders and post-only limit orders before processing good-till-cancel and immediate-or-cancel orders. The rationale is grounded in microstructure theory — when prices move, market makers want to cancel stale quotes; if a faster taker is processed first, the maker is picked off. Cancel prioritization eliminates this specific source of adverse selection on resting orders by giving the maker a structural ordering advantage.

On October 10, 2025, the cryptocurrency derivatives market experienced a system-wide liquidation cascade. Across centralized and decentralized perpetual venues, leveraged positions were liquidated at unusual scale; aggregate liquidation volume exceeded \$19 billion within a 24-hour window. The cascade has been the subject of subsequent industry commentary and a formal analysis of the protocol's automated deleveraging mechanism (Chitra et al., 2025). What has not been documented is the inside-market response on Hyperliquid in detail — how quickly spreads widened, how quickly they recovered, and whether the recovery was symmetric between quoted prices and posted sizes.

This paper provides that documentation. We use Hyperliquid's public L2 order-book archive to construct a minute-level panel of inside-market metrics for the five most-traded perpetual contracts (BTC, ETH, SOL, XRP, and DOGE) across a 92-day window centered on the October 10 event. The dataset comprises 531,305 minute observations. Because the entire analysis is venue-internal, all comparisons are against the venue's own pre-event baseline; no cross-venue spread inference is required.

Three findings emerge.

First, the cascade was a tightly compressed intraday event rather than a multi-day breakdown. Hour-by-hour analysis of October 10 UTC shows the inside spread remaining at normal levels (0.37 to 0.44 basis points pooled across all five assets) through hour 14, beginning to widen at hour 15, elevating moderately through hours 16 to 20, then peaking at hour 21 (median 9.38 bps, p95 87.48 bps, maximum 241.8 bps). By hour 22 median spread was already down to 1.71 bps, and by hour 23 to 1.08 bps. Realized volatility followed the same compressed timeline. The bulk of the venue's stress event was contained within a three-hour window.

Second, spread recovery across the 23-day window between the cascade and a separate market-wide selloff beginning November 3, 2025, was close to complete. Taking daily medians at the end of the clean post-event window (November 2, 2025) relative to pre-event baseline (October 3-9, 2025), the five assets returned to between 1.06 and 1.19 times pre-event spread, with three of five

within 1.1 times. The fastest recoveries (BTC 1.07, DOGE 1.06) reached essentially pre-event levels; the two laggards (SOL 1.19, XRP 1.14) reached within approximately 20 percent. A separate market-wide selloff beginning November 3-4 pushed the spread trajectory back into the 1.3 to 1.5 range for all assets, and a subset of assets did not return to within 10 percent of pre-event spread by the end of the full 66-day observation window; we treat the November 3-4 event explicitly in Section 6 and show that the clean-window recovery finding is robust to its exclusion.

Third, inside-quote depth contracted persistently over the same clean window, even as spreads recovered. Four of five contracts showed further depth contraction from the first post-event week to the third post-event week, with end-of-window depth ratios between 0.66 and 0.79 of pre-event levels. Two contracts (BTC and DOGE) exhibited depth contractions of more than 15 percentage points over this window. This contraction preceded the November 3-4 selloff and is therefore attributable to the cascade itself, not to the subsequent event. Market makers maintained competitive quoted prices but reduced the inventory they were willing to commit at those quotes.

The contrast between the two findings is what we term two-regime recovery: pricing recovered quickly and nearly completely within the clean post-event window, while inventory commitment contracted persistently over the same window. This pattern has two implications. For microstructure theory, it is consistent with the inventory-management framework (Stoll, 1978; Ho and Stoll, 1981), in which a tail event raises the perceived risk of holding inventory and the natural maker response is size reduction rather than price markup. For execution-quality measurement, it implies that spread-only metrics systematically understate the post-event impairment of venue liquidity, because depth has deteriorated more and more persistently than spread.

These findings contribute to two literatures. First, to the empirical microstructure of decentralized derivatives venues, which until recently has been data-constrained. The public archive that Hyperliquid publishes makes inside-market analysis at this granularity feasible from outside the venue — we use only freely available data with approximately ten US dollars of egress cost — and provides a methodological template for replication on other on-chain venues with similar archive practices. Second, to the literature on market resilience after stress events, where almost all empirical work has focused on equity venues (Brogaard et al., 2018; Anand and Venkataraman, 2016; Bessembinder et al., 2016). We provide what we believe is the first detailed inside-market resilience study on a major perpetual futures venue, centralized or decentralized.

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the institutional setting. Section 4 details the data and sample construction. Section 5 characterizes the

venue's normal-state inside market. Section 6 presents the cascade event study with the truncated-window robustness check. Section 7 discusses interpretation and limitations. Section 8 concludes.

2. Related Literature

2.1 Cryptocurrency Market Microstructure

The microstructure of cryptocurrency markets has been an active research area since the emergence of derivatives trading on these assets. Foley, Karlsen, and Putnins (2019) document order-flow dynamics on early Bitcoin spot exchanges. Brauneis, Mestel, Riordan, and Theissen (2021) provide a comprehensive treatment of liquidity measurement in cryptocurrency markets, comparing transaction-cost proxies against high-frequency benchmarks across major exchanges. He and Manela (2024) study perpetual-spot arbitrage and the associated execution costs. Cong, Landsman, and Maydew (2024) analyze how the introduction and removal of perpetual contracts affects spot-market liquidity and price efficiency. None of these papers analyze inside-market behavior on a decentralized perpetual venue, which is the focus of our work.

On-chain venues introduce distinct microstructure features absent from centralized exchanges: deterministic block-level execution, validator co-location, and protocol-defined order priorities. Chitra, Angeris, et al. (2025) provide the first formal model of automated deleveraging mechanisms for perpetual exchanges and analyze them empirically using the Hyperliquid October 10 cascade. Their work establishes a trilemma between solvency, revenue, and trader fairness in the design of forced-liquidation mechanisms. Our paper is complementary: it analyzes the inside-market response to the same event from the perspective of voluntary market-maker behavior rather than protocol-enforced liquidations.

2.2 Market Resilience After Stress Events

A substantial literature studies how equity markets respond to and recover from stress events. Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018) document liquidity dynamics around extreme price moves on US equity venues, finding that high-frequency traders supply liquidity during ordinary stress but withdraw during extreme stress. Anand and Venkataraman (2016) study recovery dynamics after market-wide circuit-breaker triggers. Bessembinder, Hao, and Zheng (2016) examine the persistence of spread widening following corporate-event-driven volatility. The shared finding across these papers is that recovery is asymmetric: spreads widen quickly in stress and narrow more slowly, with the slowness driven by maker risk-aversion rather than by underlying volatility persistence.

A common limitation of this literature is that it focuses almost exclusively on the spread dimension of execution quality, with depth treated as a secondary consideration or omitted entirely. Our two-regime finding suggests this focus misses the more persistent dimension of post-event impairment in at least one class of venue. To our knowledge no paper in this literature has examined a perpetual futures venue, and no paper has examined a decentralized venue. The Hyperliquid October 10 cascade is well-suited to extending this literature because it occurred on a venue with continuous (24/7) trading and no circuit breakers, providing a clean observation of maker-driven recovery dynamics absent venue-imposed interventions.

2.3 Inventory Management and Maker Responses to Risk

The theoretical foundation for our two-regime interpretation is the inventory-management framework of Stoll (1978) and Ho and Stoll (1981), in which market makers set quotes as a function of both expected adverse selection and the cost of holding inventory exposed to price risk. In this framework, a tail event that raises the perceived variance of inventory can rationally produce a separable response: competitive quotes are maintained to capture order flow, but the size at those quotes is reduced to limit inventory accumulation. The two-regime pattern we document is consistent with this model. The literature has previously documented the model's predictions in other settings (e.g., Madhavan and Smidt, 1991; Hasbrouck and Sofianos, 1993), but we are not aware of a direct decomposition of post-event recovery into pricing and inventory dimensions separately for a major perpetual futures venue.

2.4 Cancel Prioritization and Maker Protection

The literature on order-priority rules is long but has been relatively quiet on cancel prioritization specifically. Hendershott and Moulton (2011) document that the introduction of NYSE's Hybrid Market — which reduced the time-priority advantage of slow-moving liquidity providers — measurably worsened execution quality for retail-sized orders, suggesting that protections for resting orders matter for inside-market quality. Foucault, Kadan, and Kandel (2013) provide a theoretical treatment of cancel-related fees and their effect on quote-update behavior. Hyperliquid's design implements a particularly aggressive form of cancel prioritization: at the consensus layer, all cancels are processed before any matching, regardless of submission time. To our knowledge no centralized exchange operates this rule in its full form.

3. Institutional Setting

3.1 Hyperliquid

Hyperliquid is a layer-one blockchain whose primary application is a perpetual-futures decentralized exchange. The protocol uses HyperBFT, a modified HotStuff consensus algorithm with sub-second block times. All order-book operations — placement, modification, cancellation, and matching — execute on-chain within consensus blocks. Validators are tightly co-located, which keeps consensus latency low enough to support order-book trading workflows familiar from centralized exchanges. Trading is fee-funded rather than gas-funded, eliminating one source of execution variance present on other on-chain venues.

The protocol's matching rules differ from a standard continuous limit order book in one substantive way: within each block, the validator software is required to process cancel orders and post-only limit orders before processing good-till-cancel and immediate-or-cancel orders. The reasoning is grounded in standard microstructure theory. When the underlying spot price moves, a market maker with resting quotes faces an immediate adverse-selection threat: any taker arriving in the same block is more likely to be informed about the price move than the average trader. The maker's natural response is to cancel and re-quote. Without cancel prioritization, a sufficiently fast taker can submit a marketable order in the same block as the maker's cancel and have it processed first — the maker is picked off at a stale price. Cancel prioritization eliminates this specific failure mode.

The Hyperliquid Foundation publishes a daily archive of L2 order-book snapshots to a public S3 bucket ([s3://hyperliquid-archive](https://s3.amazonaws.com/hyperliquid-archive)). As measured in our sample, snapshots are produced at a median inter-snapshot gap of 0.543 seconds, yielding approximately 110 snapshots per asset per minute. Each snapshot contains the top 20 levels on each side of the book. Trade data are available separately via the Hyperliquid API but are not included in the S3 archive; the analysis in this paper uses only book-snapshot data and does not require trade-level inputs.

3.2 The October 10, 2025 Liquidation Cascade and the November 3-4 Selloff

On October 10, 2025, the cryptocurrency perpetual futures market experienced a system-wide liquidation cascade. Across centralized and decentralized venues, leveraged positions were forcibly closed at scale; aggregate liquidation volume across all major venues exceeded \$19 billion within a 24-hour window. Hyperliquid's automated deleveraging mechanism was triggered for several contracts, and the venue's inside market widened sharply for a multi-hour window beginning approximately 21:00 UTC. Chitra et al. (2025) provide a formal analysis of the

protocol's response from the perspective of forced-liquidation mechanism design. The present paper analyzes the same event from the complementary perspective of inside-market dynamics.

Twenty-five days after the cascade, on November 3-4, 2025, the market experienced a smaller but still substantial selloff. Bitcoin declined from approximately \$107,000 on November 1 to \$99,000-\$104,000 on November 4, with daily moves in excess of 5 percent and renewed liquidations across major perpetual venues. The selloff was attributed to a combination of macroeconomic risk-off positioning related to AI-trade concerns in equity markets and renewed crypto-specific deleveraging. The November event is an order of magnitude smaller than October 10 in absolute liquidation volume but is large enough to produce visible widening in our post-event recovery trajectory. We treat the cascade as the primary event and the November selloff as a confound to be explicitly controlled for in the empirical analysis.

4. Data

4.1 Sample Construction

Our sample covers September 15 through December 15, 2025, a 92-day window centered on the cascade event. We restrict the asset universe to the five perpetual contracts with continuous existence on Hyperliquid throughout the sample and with sufficient daily volume to support meaningful inside-market inference: BTC, ETH, SOL, XRP, and DOGE.

4.2 Hyperliquid Order-Book Data

We obtain Hyperliquid L2 order-book snapshots from the public archive bucket. The bucket is requester-pays, meaning the data are free at source but the requester bears the egress cost; total data-transfer cost for our sample was approximately ten US dollars. Snapshots are stored as line-delimited JSON within hourly LZ4-compressed files. We extracted, parsed, and re-stored each asset-day as a Parquet file, retaining the original timestamps and the full twenty-level book on each side.

From each L2 snapshot we compute the quoted half-spread at the inside (in basis points relative to the midpoint), the inside-quote depth in US dollars (the dollar size resting at the best bid plus the dollar size resting at the best ask), and the simulated round-trip execution cost for marketable orders at \$50K, \$100K, and \$500K notional sizes. Realized volatility is computed as the annualized standard deviation of one-second log midpoint returns within each minute, expressed in basis points. All metrics are aggregated to one-minute resolution by taking the first value within each minute bin.

4.3 Sample Sizes and Coverage

After construction, the venue-internal panel contains 531,305 minute observations across 78 dates and 5 assets. Coverage is essentially complete in the pre-event baseline window (October 3-9, 2025: 10,080 minutes per asset) and in the primary post-event window (October 11 through November 2, 2025: approximately 25,000 minutes per asset accounting for a six-day archive gap October 17-22). The Hyperliquid public archive also published no L2 snapshots for November 18-25, 2025; observations for those date ranges are missing from the extended recovery analysis.

5. Pre-Event Inside-Market Characterization

Before turning to the event study, we characterize the venue's normal-state inside market over the pre-event baseline window of October 3-9, 2025.

Table 1. Pre-event inside-market baselines (October 3-9, 2025).

Asset	Median spread (bps)	Mean spread (bps)	Median depth (USD)	Median realized vol (bps)
BTC	0.088	0.097	1,825,661	2.11
ETH	0.238	0.248	1,400,559	3.65
SOL	0.467	0.479	444,653	5.57
XRP	0.369	0.385	77,696	4.49
DOGE	0.562	0.602	28,135	6.46

Note: Each cell is computed across 10,080 minute observations per asset (7 days $\times$ 1,440 minutes). Inside depth is the dollar value resting at the best bid plus the best ask, summed across both sides. Realized volatility is the annualized standard deviation of one-second log midpoint returns within each minute, in basis points.

The cross-sectional pattern is consistent with stylized facts about market quality across asset capitalizations. BTC quotes a tight inside spread (median 0.088 bps) backed by approximately \$1.8 million of inside depth; DOGE quotes a wider spread (0.562 bps) backed by approximately \$28,000 of inside depth. The five assets span roughly an order of magnitude in pre-event spread and approximately three orders of magnitude in pre-event depth.

5.1 Intra-Day and Day-of-Week Patterns

On a 24/7 venue with no opening or closing auctions, we find essentially flat intra-day and day-of-week patterns in the median quoted spread pooled across assets. Across all 24 hours of the UTC trading day, the median spread varies between 0.437 and 0.471 basis points — a 7 percent range. Across all seven days of the week, the median varies between 0.431 and 0.457 basis points — a 6

percent range. This contrasts with conventional equity venues, where intra-day patterns of spread and depth are well-documented (Wood, McInish, and Ord, 1985), and with traditional futures markets, where overnight and weekend gaps produce significant time-of-day effects. The flatness on Hyperliquid is itself an institutional feature of decentralized perpetual venues: continuous global trading without cohort handovers produces a stationary inside market in normal state.

6. The Cascade Event Study

6.1 Intraday Evolution on October 10

Figure 1 presents the minute-level evolution of inside spread on October 10, 2025 UTC, for each of the five assets. Spreads on all five assets remained near baseline through approximately hour 13. A first elevation is visible around hour 15 (corresponding to a brief BTC drawdown), partial recovery through hour 17, and a second elevation around hour 19. The principal cascade peak occurred in hour 21 UTC, with all five assets simultaneously experiencing spread widening of one to two orders of magnitude. By the end of the trading day all assets had partially recovered toward normal levels, although none had returned to pre-event baselines.

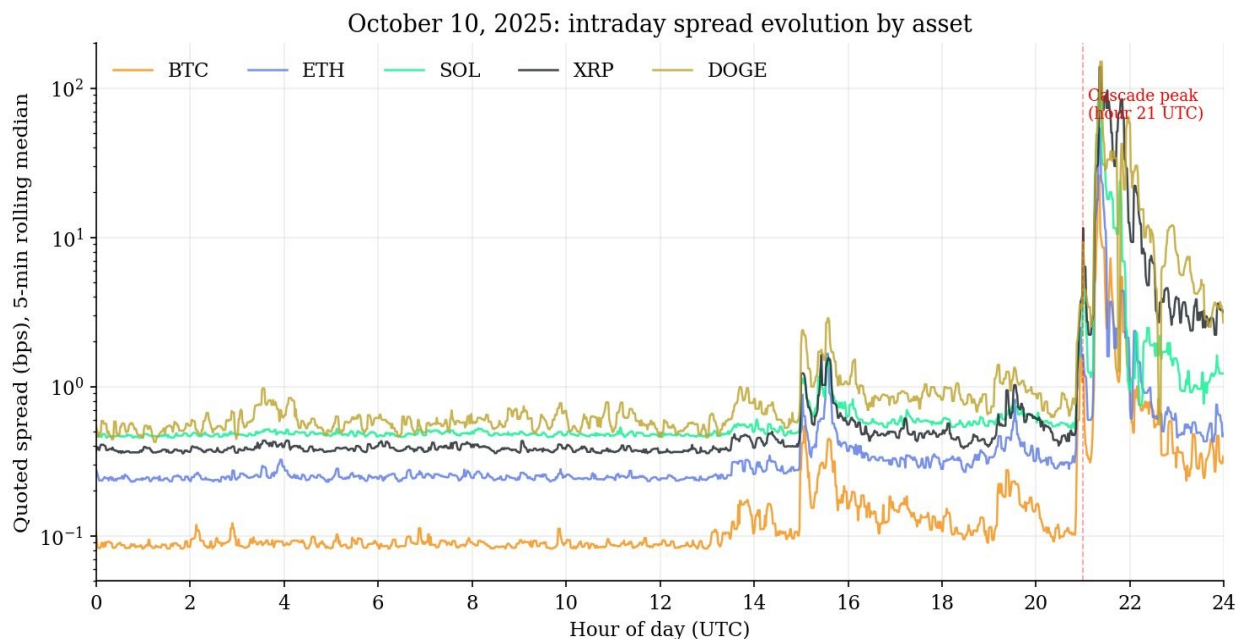


Figure 1. Minute-level inside spread on October 10, 2025 UTC, by asset. Five-minute rolling median, log scale. Vertical dashed line indicates the principal cascade peak at hour 21 UTC. The first 14 hours of the day are essentially indistinguishable from a normal trading day; the cascade is concentrated in a three-hour window from approximately hour 21 to hour 23.

Table 2 reports the hour-by-hour evolution pooled across assets, providing quantitative anchors for the qualitative pattern in Figure 1.

Table 2. Hour-by-hour inside-market evolution, October 10, 2025 UTC.

Hour UTC	Median spread (bps)	p95 spread (bps)	Max spread (bps)	Median depth (USD)	Median rv (bps)
00	0.372	0.566	0.639	415,716	3.43
04	0.378	0.667	1.014	604,670	4.59
08	0.381	0.658	0.904	520,120	4.37
12	0.368	0.577	0.728	493,769	3.44
13	0.399	0.802	1.071	548,614	6.64
14	0.441	0.802	4.135	597,182	7.27
15	0.724	1.983	4.968	463,643	16.60
16	0.497	1.009	1.568	623,944	9.89
17	0.485	0.936	1.188	487,329	8.99
18	0.439	0.960	1.178	515,262	8.04
19	0.588	1.291	2.852	566,483	12.78
20	0.534	3.030	13.692	514,913	8.66
21	9.376	87.476	241.819	391,473	150.62
22	1.709	18.804	44.351	375,291	42.46
23	1.080	5.576	8.713	217,889	31.14

Note: Statistics are pooled across all five assets within each UTC hour bin. Hours 1, 2, 3, 5, 6, 7, 9, 10, 11 are similar to hour 0 and omitted for brevity.

The first 14 hours of the day were essentially indistinguishable from a normal trading day. Beginning hour 15, both spread and realized volatility elevated noticeably; through hours 16 to 20, the inside market was moderately stressed. The peak occurred in hour 21: median spread reached 9.38 bps (more than 25 times normal levels), the 95th percentile reached 87.5 bps, the maximum minute observation reached 241.8 bps, and median realized volatility reached 150.6 bps (more than 30 times normal). By hour 22 the median spread had already declined to 1.71 bps, and by hour 23 to 1.08 bps. Inside depth contracted modestly during the peak hour and continued to drop into hour 23 (\$218K).

6.2 Spread Recovery in the Clean Post-Event Window

Because a separate market-wide selloff began on November 3, 2025, our primary recovery analysis focuses on the 23-day clean window between the cascade and the second event: October 11

through November 2, 2025. Table 3 reports the key recovery metrics in this window, computed from daily medians relative to the pre-event baseline.

Table 3. Recovery metrics in the clean post-event window (October 11 – November 2, 2025).

Asset	Max spread ratio	End-of-window spread ratio	Days to $\leq 1.2\times$	Days to $\leq 1.1\times$
BTC	1.31 $\times$	1.07 $\times$	6	15
ETH	1.41 $\times$	1.13 $\times$	6	15
SOL	1.43 $\times$	1.19 $\times$	14	Not reached
XRP	1.82 $\times$	1.14 $\times$	15	Not reached
DOGE	1.77 $\times$	1.06 $\times$	13	14

Note: Spread ratio is the daily median quoted spread relative to the pre-event baseline (October 3-9, 2025). "Max" refers to the maximum daily median observed during the 23-day window. "End-of-window" is the spread ratio on November 2, 2025. Threshold-recovery columns report the number of days after October 10 required to first reach the threshold; "Not reached" indicates the threshold was not crossed within the 23-day window.

The recovery in quoted spread is rapid and close to complete. Within 6 days, BTC and ETH were within 1.2 times pre-event spread; within 13-15 days, BTC, ETH, and DOGE were within 1.1 times. By the end of the 23-day window, BTC (1.07 $\times$) and DOGE (1.06 $\times$) were essentially fully recovered, ETH (1.13 $\times$) was close, and the two laggards (SOL at 1.19 $\times$ and XRP at 1.14 $\times$) were within approximately 20 percent. This pattern contrasts with the more common stylized finding that stress-event spread widening persists for extended periods; on Hyperliquid, the spread dimension of execution quality was substantially restored within a month of a major cascade.

Figure 2 presents the seven-day rolling spread ratio for each asset across the full 51-day post-event window. The clean-window recovery documented in Table 3 is visible in days 1-23. Beyond day 23, the trajectory deviates upward coincident with the November 3-4 selloff, producing a second wave of widening that does not reflect Hyperliquid-specific dynamics but rather a market-wide event superimposed on a venue in partial recovery.

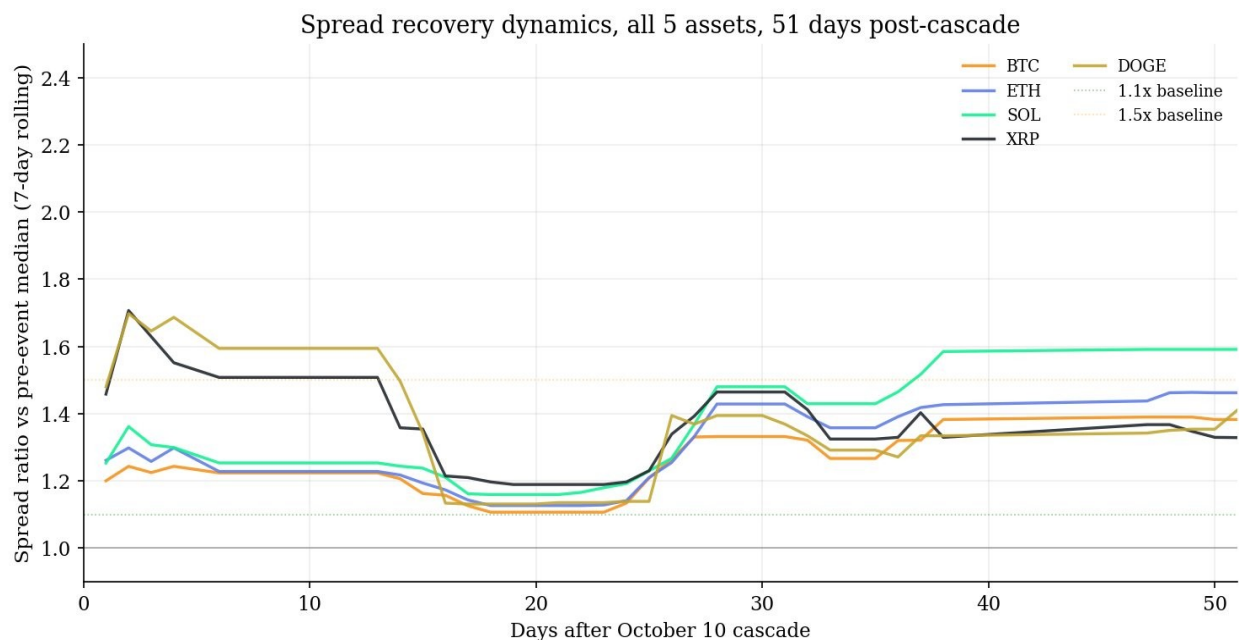


Figure 2. Daily spread ratio relative to pre-event baseline, seven-day rolling median, by asset. Days 1-23 correspond to the clean post-event window (October 11 - November 2). Days 24 onward are confounded by the November 3-4 market-wide selloff.

6.3 Depth Contraction: The Persistent Dimension

The recovery story in depth is qualitatively different from the spread story. Table 4 reports depth dynamics in the same clean post-event window.

Table 4. Depth contraction in the clean post-event window (October 11 – November 2, 2025).

Asset	Days 1-7 median depth ratio	Days 17-23 median depth ratio	Change (percentage points)
BTC	0.93×	0.77×	-16.2
ETH	0.88×	0.79×	-9.8
SOL	0.93×	0.90×	-2.7
XRP	0.76×	0.78×	+1.5
DOGE	0.97×	0.83×	-14.7

Note: Depth ratio is the daily median inside-quote depth relative to the pre-event baseline. "Days 1-7 median" aggregates October 11-16 (with October 17 missing due to archive gap). "Days 17-23 median" aggregates October 27 - November 2.

Four of five assets (all except XRP) showed further depth contraction from the first week to the third week of the clean post-event window. BTC depth declined by 16.2 percentage points, DOGE

by 14.7 percentage points, ETH by 9.8 percentage points, SOL by 2.7 percentage points. Only XRP showed any depth recovery, and the improvement was marginal (+1.5 pp) from an already-low starting level (0.76 $\times$). All assets ended the clean window with depth below 0.80 $\times$ of pre-event levels. The depth contraction preceded the November 3-4 event and is therefore attributable to the cascade itself, not to the subsequent disturbance.

Figure 3 extends the depth analysis through the full 51-day post-event window. The pattern differs markedly from the spread figure: depth trajectories are monotonically declining through the entire window with no visible second-event signature. Two contracts (ETH and DOGE) cross the 0.5 $\times$ line by the end of the observation period — inside depth cut by half relative to pre-event seven weeks later.

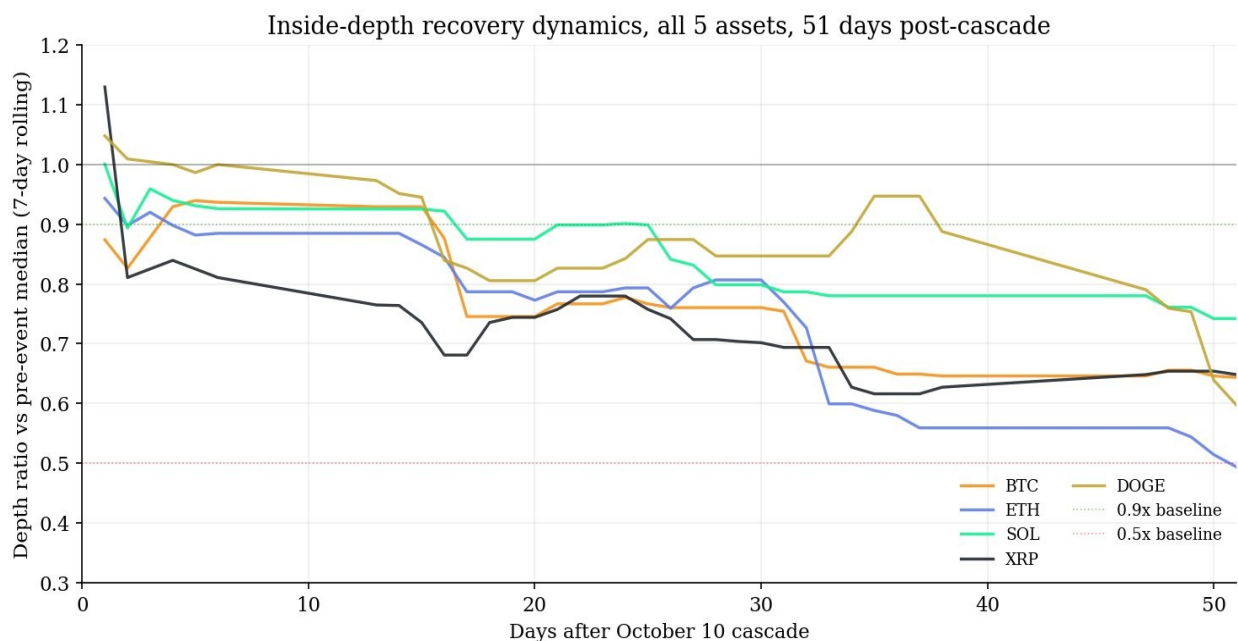


Figure 3. Daily depth ratio relative to pre-event baseline, seven-day rolling median, by asset. Unlike the spread series in Figure 2, depth shows monotonic deterioration over the entire 51-day window, with no visible second-event inflection. Reference lines at 0.9 $\times$ (modest contraction) and 0.5 $\times$ (severe contraction).

6.4 Cross-Sectional Pattern

Figure 4 presents the cross-sectional pattern at +30 days post-cascade. The left panel plots spread widening against pre-event spread (a proxy for asset liquidity rank); the right panel plots depth contraction against the same.

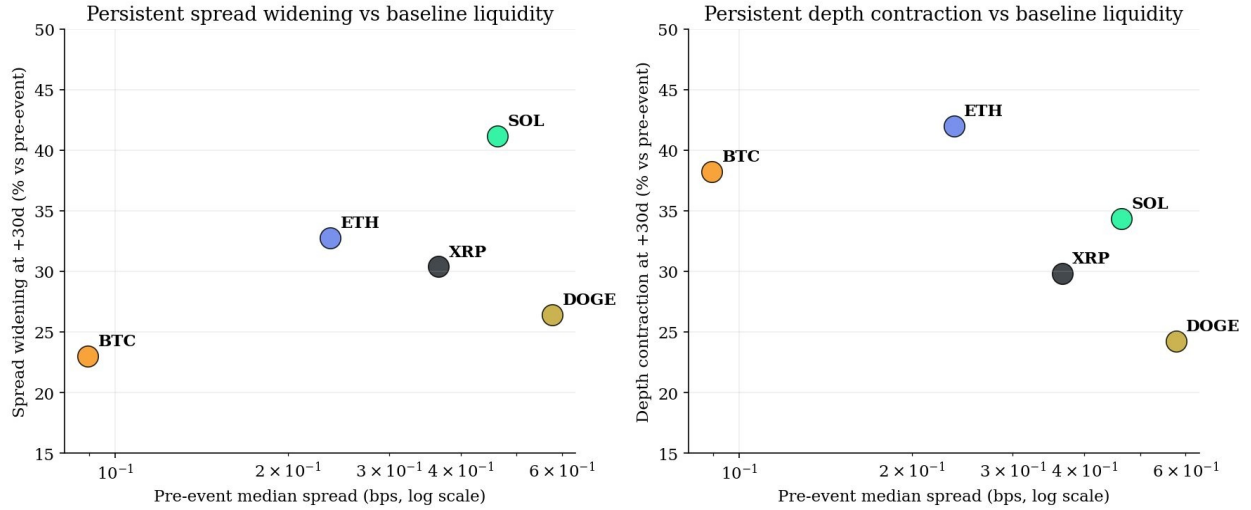


Figure 4. Cross-sectional pattern of persistent post-event effects at +30 days. Left: spread widening (% vs pre-event). Right: depth contraction (% vs pre-event). The +30 day horizon falls after the November 3-4 event, so the spread-panel numbers include second-event effects; the depth-panel numbers remain attributable to the cascade per the robustness analysis in Section 6.3.

The depth cross-section is the cleaner finding and is interpretable as follows. BTC and ETH (the two largest contracts by pre-event depth) exhibit the largest depth contractions (38% and 42%). The pattern is consistent with an interpretation in which tail-risk perception increases proportionally to notional exposure: makers committed the most inventory to the largest contracts pre-event and therefore had the most inventory to reduce post-event. The smallest contract (DOGE) exhibits the smallest depth contraction, but this should not be interpreted as DOGE being least affected — DOGE had the least depth to begin with, and the absolute reduction in inventory commitment is larger for the larger contracts.

7. Discussion

7.1 Two-Regime Recovery

The central empirical finding of this paper is the two-regime recovery pattern: pricing recovers quickly and nearly completely within three weeks of a cascade, while inventory commitment contracts persistently over the same window and continues to deteriorate over the full observation period. The pattern is consistent with the inventory-management framework of Stoll (1978) and Ho and Stoll (1981), in which market makers set both quoted prices and committed sizes as separable decisions responding to different risk considerations. A tail event raises the perceived variance of inventory holdings, and the rational maker response is to reduce size exposure rather than demand a wider spread — particularly in a competitive market where quoted prices are disciplined by the possibility of being undercut.

The two-regime pattern has implications for how venue resilience should be measured. A metric that examines only the spread dimension — as is conventional in most empirical microstructure work — will systematically under-state the post-event impairment of venue liquidity when makers respond primarily through size reduction. A complete resilience assessment requires monitoring both dimensions, with particular attention to the depth dimension which our evidence suggests is the more persistent and potentially more consequential for large-order execution quality.

7.2 Implications for Cancel Prioritization

Hyperliquid's distinguishing market-design feature is the consensus-layer cancel prioritization rule, motivated by maker protection. Our findings are consistent with this protection operating during normal-state trading: pre-event spreads are tight relative to the venue's age, and intra-day patterns are flat (Section 5). The rapid spread recovery we document (Section 6.2) is also consistent with an environment where makers feel confident re-entering quoted prices quickly post-stress. But the persistent depth contraction (Section 6.3) suggests that cancel prioritization, while beneficial for normal-state market quality and for fast pricing recovery, does not insulate the venue from the inventory-commitment effects of a tail event. The protection that cancel prioritization offers is against being picked off on individual quotes; it does not reduce the underlying variance of crypto-perpetual inventory across a market-wide stress event. Venue resilience and venue-design quality are distinct properties.

7.3 Limitations

We list the principal limitations. First, the Hyperliquid public archive publishes only the top twenty levels of the book on each side. Wider-book dynamics (depth at, say, 25 basis points outside the inside) are not directly observable from the public data; the depth metric we report captures top-of-book commitment specifically. Second, the archive published no L2 snapshots for October 17-22 and November 18-25, 2025. The first gap falls within the clean post-event window and reduces our coverage of the 7-12 day post-cascade range; the second gap falls in the extended post-event window. Third, we cannot directly identify individual market makers; our analysis is silent on whether the persistent inventory contraction reflects a few large makers exiting or many small makers each reducing exposure marginally. Fourth, the November 3-4 selloff confounds post-event analysis beyond day 23; our primary recovery findings are restricted to the clean window, but the extended recovery discussion in Figure 2 and in Section 6.4's spread panel incorporates both events. Fifth, our analysis is venue-internal by design and does not address whether Hyperliquid's recovery dynamics are typical or atypical relative to centralized perpetual venues experiencing the same cascade.

8. Conclusion

This paper documents the inside-market response of Hyperliquid — the largest decentralized perpetual futures venue — to the October 10, 2025 system-wide liquidation cascade. Using the venue's public L2 archive, we construct a minute-level panel spanning five major perpetual contracts across a 92-day window centered on the event. Three findings emerge.

First, the cascade was a tightly compressed intraday event. Median quoted spread pooled across assets reached 9.4 basis points at hour 21 UTC and the 95th percentile reached 87.5 basis points, with intraday recovery well under way by end of day. Second, in the 23-day clean post-event window prior to a separate market-wide selloff, quoted spreads recovered substantially — three of five contracts within 1.1 times pre-event levels and all within 1.2 times. Third, inside-quote depth contracted persistently over the same clean window, with four of five contracts showing further depth reduction from days 1-7 to days 17-23 post-event and none fully recovering over the full 51-day observation window.

The contrast between the two findings — spread recovery near complete while depth contraction persists — motivates what we term two-regime liquidity recovery. Market makers maintain competitive quoted prices but reduce the inventory they commit at those quotes following a tail event. Three implications follow. For microstructure theory, the finding is consistent with the classic inventory-management framework's prediction that pricing and sizing are separable maker decisions responding to different risk considerations. For empirical market-resilience research, the finding implies that spread-only metrics systematically understate the impairment of venue liquidity following stress events and that depth metrics should be a standard component of resilience analysis. For decentralized derivatives venue design, the finding establishes that a venue specifically engineered to protect quoting market makers can still experience multi-week impairment of inventory commitment after a tail event; venue resilience and venue-design quality are distinct properties.

Beyond the specific findings, this paper takes a methodological position: the public order-book archives that on-chain venues publish offer opportunities for transparent, replicable empirical microstructure research that are difficult to match in centralized venues that typically restrict tick-level access. The dataset used in this paper was assembled at approximately ten US dollars of total cost. We hope the results encourage future replication and extension on other on-chain venues with similar archive practices.

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